

JEE MAIN REVISION MASTER: SALT ANALYSIS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 4 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Systematic Qualitative Inorganic Salt Analysis	4 PYQs	Inorganic Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Systematic Qualitative Inorganic Salt Analysis	$\text{Brown Ring Complex: } [\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+} \setminus (\text{Fe has } +1 \text{ oxidation state})$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Systematic Qualitative Inorganic Salt Analysis (Moderate Yield — 4 PYQs)

Core Principle: Group reagents for cations: Group I (Pb^{2+} , dil. HCl), Group II (Cu^{2+} , Pb^{2+} , H_2S in acidic medium), Group III (Fe^{3+} , Al^{3+} , NH_4OH in presence of NH_4Cl), Group IV (Zn^{2+} , Mn^{2+} , Ni^{2+} , H_2S in ammoniacal medium). Anion confirmatory tests: Brown ring test for nitrate (NO_3^-) forms $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$. Nessler's reagent ($\text{K}_2[\text{HgI}_4]$) gives brown precipitate with ammonium salts.

■ **Caution / JEE Pitfall:** In the brown ring complex $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$, iron has an anomalous oxidation state of +1 (NO acts as neutral/positive NO^+ ligand).

■ **Speed Trick:** Group III cations require NH_4Cl before NH_4OH to suppress OH^- concentration via common ion effect, preventing precipitation of Group IV hydroxides.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "When salt BA is treated with Conc. H_2SO_4 reddish brown gas is liberated. The aqueous solution of BA gives pale yellow ..." [Ans: D]
- [2025 Shift] Q#2 (MCQ): "The group reagent $\text{NH}_4\text{Cl(s)}$ and aqueous NH_3 will precipitate which of the following ion?..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The sodium fusion extract is boiled with concentrated nitric acid while testing for halogens. By doing so, it..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

When salt BA is treated with Conc. H_2SO_4 reddish brown gas is liberated. The aqueous solution of BA gives pale yellow precipitate with AgNO_3 solution. Which of the following anion (A^-) is present in the salt BA?

- (A) Cl^-
- (B) CO_3^{2-}
- (C) SO_4^{2-}
- (D) Br^- [Correct]

Correct Answer: D

Solution Analysis: The anion present is Br^- . When a bromide salt is treated with concentrated H_2SO_4 , reddish brown vapours of bromine are liberated. First, hydrogen bromide is formed: $\text{BBr} + \text{H}_2\text{SO}_4 \rightarrow \text{BHSO}_4 + \text{HBr}$. Then HBr reduces concentrated H_2SO_4 and gets oxidised to bromine: $2\text{HBr} + \text{H}_2\text{SO}_4 \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$. The bromine vapours, Br_2 , are reddish brown in colour. Also, the aqueous solution of the salt gives a pale ye